OPTIONAL CHALLENGE– UART TRANSMITTER

1. Design Approach:

Designed a UART transmitter module using Verilog HDL to convert 8-bit parallel data into serial data transmission. Implemented a finite state machine consisting of four states: IDLE, START, DATA and STOP. Used a clock counter to control the duration of each transmitted bit and a transmission-done signal to indicate completion.

2. Verification Steps:

- Created a separate testbench.
- Applied reset to initialize the UART transmitter.
- Provided an 8-bit input data value.
- Activated the transmission-start signal.
- Observed the serial TX output.
- Checked the start bit, eight data bits and stop bit.
- Verified the transmission-done signal.
- Simulated using ModelSim.

3. Results:

The testbench checks serial transmission of the input data 10100101. The UART transmitter sends one start bit, eight data bits in LSB-first order and one stop bit. The transmission-done signal indicates completion of data transmission.

4. Issues Faced:

Implementing the UART state machine and maintaining correct bit timing were challenging. Proper synchronization of the clock counter is important. Correct transmission order of data bits and generation of start and stop bits must also be ensured.

5. Conclusion:

Successfully designed a UART transmitter using Verilog HDL to demonstrate parallel-to-serial data conversion and serial communication.